

What is the use of HV or VH polarization in Radar?

Asked 4 years, 3 months ago Modified 4 years, 3 months ago Viewed 2k times

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Radar is a common tool in remote sensing. It is a so-called active instrument, hence it emits and receives energy. Modern Syntethic Aperture Radar (SAR) systems are capable of both transmitting and receiving polarized waves. The most common polarizations are horizontal (HH) and vertical (VV) polarization, although other ones exist. Using H and V leads to 4 possible combinations of emitting and receiving signals:

$$HH, VV, HV, VH,$$

which is read as transmitted first, received later (i.e. HV means transmitting horizontally polarized waves and receiving only vertical polarized waves).

I can relatively easily imagine the use of the HH and VV copolarized waves. They should be easier reflected by object that have a longish geometric structure. Large VV and small HH would probably indicate tower-like structures and small VV and large HH should indicate bridge-like structures.

However it is not obvious to me what structure would reflect a cross-polarized wave HV or VH stronger than HH or VV ? Furthermore, what is the difference between an object that reflects VH and an object that reflects HV ?

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edited Mar 17, 2020 at 23:36

asked Mar 9, 2020 at 19:18



ckrk

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cool question!! – uhoh Mar 10, 2020 at 13:19

1 @uhoh thank you, i think its not that obvious! My collaborator seemed to use it in a very empirical fasion (ie trial and error of all possibilities) and couldnt give an intuition. I even checked a remote sensing textbook and it did not answer the question, despite mentioning the existence of the HV and VH polarizations.... – ckrk Mar 10, 2020 at 16:17

1 Answer

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In the case of a ground looking radar, the cross-polarized signals HV and VH generally increase with surface roughness. For example, dense vegetation is one of the more depolarizing surface types because the polarization orientation of the emitted beam is scrambled over the course of multiple scattering interactions within the canopy before returning to the detector. Smoother surfaces can be thought of as having fewer (often just one) interactions and the corresponding signals are thus reflected to the detector predominately in their initial orientation (i.e. as HH or VV).

The cross-polarized signal also increases as scatters suspended in the atmosphere become less spherical in shape. This is particularly applicable in the interpretation of meteorological radar observations where it is used to estimate raindrop size, distinguish between solid and liquid precipitation and differentiate biological scatterers (e.g. flocks of birds) from hydrometers.

For a fixed target, the return signals HV and VH will have equal intensity. This is a direct result of the [principle of reciprocity](#).

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edited Mar 21, 2020 at 4:06

answered Mar 21, 2020 at 0:10



Reed Espinosa

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Thank you for this answer. it addresses 95% of my question already. Your explanation makes a lot of sense and I feel I get the idea behind cross-polarized signals now. One thing is yet remaining, im still pondering the VH & HV differences. Your answer seem to indicate that mostly HV and VH will simply be equal. However, my understanding is that a smooth tower has large HH, a rough tower has large HV. My intuition is that conversly a tower should have little VV and only slighter larger VH depending on its roughness. Hence HV and VH would not match for a tower. Would you agree? – ckrk Mar 21, 2020 at 14:04

@ckrk Take a vertical metal rod. Incoming V will efficiently drive a (vertical) current in this rod, causing the rod to re-radiate exclusively to V, producing no VH. Tilt the same rod several degrees. The tilted rod will absorb incoming V almost as efficiently but re-radiate in a direction with a small horizontal component, resulting in non-zero VH. Now input H. The rod no longer absorbs the incoming radiation efficiently but what is absorbed will be re-radiated close to vertical, predominately into HV. More complicated cases are hard to intuit but reciprocity always guarantees that HV=VH. – Reed Espinosa Mar 21, 2020 at 18:51